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6 July 1956
SUPERSEDING
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MILITARY SPECIFICATION

MOTOR-GENERATORS, 300 KW DC TO AC; AC TO DC SUBMARINE SERVICE

1. SCOPE

1.1 This specification covers 300 kilowatts (kw.) direct current (d.c.) to alternating current (a.c.); a.c. to d.c. motor-generators for Naval submarine service.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings and publications of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

FF-B-171 - Bearings, Ball, Annular (General Purpose).

MILITARY

MIL-R-22 - Resistors, Variable (Wirebound, Power Type).
MIL-C-25 - Capacitors, Fixed, Paper-Dielectric, Direct-Current (Hermetically Sealed in Metallic Cases).
MIL-P-116 - Preservation, Methods of.
JAN-H-194 - Heaters, Electric, Air (Ship Type).
JAN-W-583 - Wire, Magnet.
MIL-S-901 - Shockproof Equipment, Class HI (High Impact) Shipboard Application, Test for.
MIL-E-917 - Electric Power Equipment, Basic Requirements for (Naval Shipboard Use).
MIL-D-963 - Drawings, Production (for Electrical and Mechanical Equipment for Naval Shipboard Use).
MIL-P-997 - Plastic-Material, Laminated, Thermosetting, Electrical-Insulating: Sheets, Glass Cloth, Silicone Resin.

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MILITARY (cont'd.)

MIL-V-1137 - Varnish, Electrical-Insulating (for Electro-Motive Equipment).
MIL-G-1140 - Glass Fiber: Yarn, Cordage, Sleeving, Cloth and Tape.
MIL-R-2033 - Relays for Naval Shipboard Electrical Service.
MIL-E-2036 - Enclosures for Electric and Electronic Equipment (Naval Shipboard Use).
MIL-C-2174 - Controllers, Motor Starters, and Master Switches - Direct-Current, Naval Shipboard Use.
MIL-C-2194 - Cables, Electric (Reduced Diameter Types) (Shipboard Use).
MIL-R-2729 - Regulator Sets, Voltage, A.C. Generator, Naval Shipboard Use.
MIL-G-3111 - Generator, Electric, Direct-Current (Naval Shipboard Use).
MIL-G-3124 - Generator, Alternating-Current, 60-Cycle (Naval Shipboard Use).
MIL-I-3505 - Insulation Sheet and Tape, Electrical, Coil and Slot, High Temperature.
MIL-P-15037 - Plastic-Material, Laminated, Thermosetting, Sheets, Glass-Cloth, Melamine-Resin.
MIL-T-15071 - Technical Manuals for Mechanical and Electrical Equipment.
MIL-R-15109 - Resistors and Rheostats, Shockproof (Naval Shipboard Use).
MIL-R-15137 - Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-S-15291 - Switches, Rotary, Snap Action, Bureau-Design.
MIL-P-16019 - Packaging of Switchboards, Controllers and Motor Starters (Over 40 Pounds).
MIL-W-16072 - Wire Magnet, High Temperature.
MIL-P-16299 - Preservation, Packaging, Packing, and Marking of Electric Machines Having Rotating Parts and Associated Repair Parts.
MIL-M-17060 - Motors; Alternating Current, Integral HP Shipboard Use.
MIL-I-17205 - Insulation Cloth and Tape; Electrical Glass-Fiber, Varnished.
MIL-M-17413 - Motors, Direct-Current, Integral HP, Naval Shipboard.
MIL-P-17721 - Plastic Sheet, Laminated, Thermosetting, Glass Mat, Melamine-Resin.
MIL-R-18281 - Rectifiers, Metallic: Selenium, Copper Oxide, and Magnesium-Copper Sulfide Types, Naval Shipboard Use.

NAVY DEPARTMENT

General Specifications for Inspection of Material.

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STANDARDS

MILITARY

MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.
MIL-STD-195 - Markings of Connections for Electric Assemblies.

DRAWINGS

BUREAU OF SHIPS

S0103-73729 - Standard Drawing Format for Production Drawings Prepared by Contractor or Manufacturer for Approval by Government Agency.
9000-S6401-74043 - Main Motor and Generator Illumination, Fixture Type A-S for Submarine.
5000-S1112-1385778 - Mounting, Resilient; EES Type.

PUBLICATIONS

BUREAU OF SHIPS

NAVSHIPS 4372 - Recommendations and Proposals for Repair Parts.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN STANDARDS

Z24.3 - Sound Level Meters for Measurement Noise and Other Sounds.
Z24.7 - Apparatus Noise Measurement.

(Application for copies should be addressed to the American Standards Association, 70 East 45th Street, New York 17, N.Y.)

3. REQUIREMENTS

3.1 General. - The equipment shall consist of a motor-generator and control equipment necessary to make a complete power supply. The set shall be capable of generating a.c. power when connected to a d.c. input power supply or d.c. power when connected to an a.c. input power supply. Detail requirements shall be as follows:

3.1.1 Enclosures. -

3.1.1.1 Motor-generator. - The motor-generator enclosure shall be water-air cooled of a double tube construction in accordance with Specification MIL-E-2036 except that no air temperature alarm will be required. The motor generator frame and accessories attached thereto shall be watertight to the bottom of the shaft opening

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and spraytight above this point. Heat exchanges shall be provided which will permit satisfactory operation when using 85° F. inlet sea water at pressures to be specified by the bureau or agency concerned. The heat exchangers shall be mounted on top and integral with the motor generators.

3.1.1.2 Control equipment. - The enclosure for motor controller and regulators shall be dripproof protected except for switch mounted components which shall be as approved by bureau or agency concerned.

3.1.2 Motor-generator frame. - The design of the motor-generator frame shall be such as to permit removal of the d.c. and a.c. machine field coils without the necessity of removing the rotor from the machine.

3.1.3 Mounting. - Each motor-generator shall be of the two bearing type with stationary components of each, the motor and generator, assembled in such a fashion that the combined unit will be capable of maintaining proper alignment without dependence upon the supporting structure of the ship, nor shall the unit require any additional structural bedplate. The rotating elements of both the motor and the generator shall be mounted on a single one piece shaft. The mounting feet shall be suitable so as to permit application of resilient mounts symmetrical about the center of gravity. The motor-generator will be mounted on eight type A6L resilient mounts in accordance with Drawing 5000-S1112-1385778. The mounting to be supplied by installing activity. Feet arrangement shall be as approved by the bureau or agency concerned. The motor controller and the regulator except components mounted on switchboards shall be suitable for back and bottom mounting.

3.1.4 Space heater. - Each motor-generator shall have an electric strip type space heater located in the lower part of the enclosure, on the commutator end. Adequate clearance (at least 2 inches) between the heater and other parts shall be provided to avoid local overheating of brush rigging, windings, and other parts of the machine. All heater terminals and all heater circuit connecting terminals and leads shall be completely protected or insulated to prevent possible contact by personnel. A warning plate shall be provided to warn personnel against servicing the machine without first deenergizing heater and lighting circuits. Provision shall be included for breaking the heater circuit internal connections when required during disassembly. Heater circuit connection terminals shall be located on an accessible terminal board protected to provide watertight cable connections. Type S heaters shall be furnished in accordance with Specification JAN-H-194. The heater shall have a capacity adequate to maintain the air temperature within the machine about 3° to 6° C. above ambient and shall be suitable for operation on 450 volt, 3 phase 60 cycle a.c. power. Heater supports shall be adequately insulated from

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the surface on which the heater is mounted to insure minimum loss of heat by conduction. The heater shall be firmly mounted to hold the heater rigidly without distortion under all operating conditions. Heater connection terminals shall terminate at the d. c. generator terminal board.

3.1.5 Internal illumination. - One or more lighting fixtures as approved by the bureau or agency concerned shall be installed for internal illumination of the commutator and brushes. Each fixture shall have a guard in accordance with Drawing 9000-S6401-74043. Lamps of the rough surface type shall be provided. The fixtures shall be located below the centerline with adequate clearance from live parts. The lighting fixtures and circuit shall be suitable for 115 volt a. c. single phase power. All terminals and connecting leads shall be completely protected or insulated to prevent possible contact by personnel. Provision shall be included for breaking the lighting circuit internal connections when required during disassembly. Lighting circuit connection terminals shall terminate at the d. c. generator terminal board.

3.1.6 Inclined operation. - The motor-generator and control equipment shall be designed for submarine application and shall be capable of satisfactory operation in accordance with the requirements of this specification when permanently inclined 30 degrees in either direction parallel to the shaft and when rolling 60 degrees to either side or with a permanent list of 15 degrees to either side.

3.1.7 Quiet operation. - The motor-generator and control equipment shall be designed to keep structural and airborne noise to a minimum. The maximum airborne noise level shall not exceed 95 decibles (db) when tested in accordance with 4.2.1.5. Noise reduction and isolation means shall include the following:

(a) Noise tested bearings.
(b) Armature slots skewed with respect to edge of pole faces.
(c) Precision balance at rated speed.
(d) End play of rotor minimized.
(e) Commutator and collector rings precision ground and all brush chatter eliminated.
(f) All electric relays, contactors and transformers shall be designed to operate quietly; all laminations shall be firmly secured to prevent chatter and magnetic hum.
(g) Ventilation fans, ducts, and vanes shall be designed to reduce aerodynamic noise to a minimum.

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3.1.8 Shock requirements. - The motor-generator and all associated control equipment shall be HI shock resistant. The hold-down bolts, feet and main structural members of the motor generator shall be designed in accordance with figure 1. The stresses caused by shock design loads acting in the vertical athwartship and fore and aft direction shall not be combined. The associated equipment shall be class HI shock resistant in accordance with Specification MIL-S-901 and all applicable equipment specifications shown in section 2.

3.1.9 Vibration. - All parts and the complete structure of the motor-generator and control equipment shall be designed to withstand externally excited vibration type I as specified in Standard MIL-STD-167.

3.1.10 Parallel operation. - Motor-generators and control equipment shall be suitable for parallel operation when generating either a. c. or d. c. power. Parallel operation shall be as follows:

(a) The a. c. generators shall operate in parallel in accordance with kv. -a. load (reactive power) division and circulating current requirements of Specification MIL-R-2729 and in accordance with the following kw. load (active power) division requirements: The actual kw. load of any generator expressed as a percentage of its continuous kw. rating shall not differ from total kw. load expressed as a percentage of the total continuous kw. rating of all connected generators by more than 10 percent. The kw. and kv. -a. load sharing requirements specified herein apply for input voltage variation within the specified limit.

(b) When the d. c. generators operate in parallel without the voltage regulators under load condition varying from 20 percent to 100 percent rated load, and varying d. c. voltage from 270 to 355 volts the difference in current supplied by any one d. c. generator from that supplied by the other generator shall not exceed 15 percent of the rated load current of the connected generators. When the d. c. generators operate in parallel with the voltage regulators under load conditions varying from 20 percent to 100 percent rated load and varying d. c. voltage from 270 to 355 volts the difference in current supplied by any one d. c. generator from that supplied by the other generator shall not exceed 7. 5 percent of the rated load current of the connected generators.

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3.1.11 Ambient temperatures. - The ambient temperature of the motor-generator and associated control equipment shall be 50°C.

3.1.12 Dimensions and weight. - The motor-generator and associated control equipment shall be as small and light as possible consistent with the performance specified herein. The mounting arrangement of motor-generator and control equipment shall be as approved by bureau or agency concerned. The length, width, height and weight of the motor-generator may be less than those specified as follows but shall not exceed those limits:

(a) Motor-generator (complete package with heat exchanger and exciter if furnished). -
Length 91 inches.
Width 72 inches.
Height 65 inches.
Weight 16,000 pounds.

(b) Motor controller. -
Width 31 inches.
Height 48 inches.
Depth 32 inches.
Weight 1000 pounds.

(c) D.c. regulator. -
Width 20 inches.
Height 42 inches.
Depth 16 inches.
Weight as approved by the bureau or agency concerned.

(d) A.c. regulator. - As approved by bureau or agency concerned.

3.1.13 Bearings. - Either rolling contact or sliding surface bearings shall be used. The lubricating system shall be self-contained and capable of providing satisfactory lubrication to bearings under the operating conditions specified in 3.1.6. The bearing housing shall be suitably sealed so as to prevent leakage of lubricant.

3.1.13.1 If rolling contact bearings are used they shall both be locked on the shaft and one bearing shall be locked in housing. The other bearing shall be floated in the housing to accommodate shaft expansion. Bearings shall be either of the prelubricated double seal, single row ball, double row width type or single shielded type in accordance with Specification FF-B-171 and shall be specially selected for quiet operation.

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3.1.13.2 If sliding contact bearings are used, a means for supporting the thrust load shall be provided.

3.1.13.3 If oil is used as the lubricant, suitable sight glasses or dip sticks shall be incorporated in the housing design for determining the level of lubricant.

3.1.13.4 If sliding contact bearings are used they shall be provided with embedded thermocouples.

3.1.14 Thermocouples. - Thermocouples of copper constantan shall be provided to measure temperature of the air entering the coolers, temperature of the air leaving the coolers, temperature of the stator winding and temperature of the bearings (if shielding contact bearings are used). The thermocouples shall be brought to terminal boards inside terminal boxes. The terminal boxes shall be suitable for installing cable clamps or stuffing tubes. The number of terminal boxes supplied shall be kept to a minimum.

3.1.15 Terminal boxes. - Terminal boxes shall be provided and securely bolted or welded to the frame. Except for thermocouples, terminal boxes shall be such that they can be mounted on either side of the motor-generator. The boxes shall not have any openings but shall be suitable for drilling and installing cable clamps or stuffing tubes by the installing activity. Gaskets shall be furnished under all terminal box covers; removable plates; and between the box and motor-generator frame, if the box is bolted to the frame.

3.1.16 Terminal boards. - All motor-generator leads shall terminate at a terminal board within terminal boxes. The terminal board shall be securely mounted within the terminal box in such a manner as to permit connection of the internal leads to the back of the board and the external leads to the front of the board. Terminal studs shall be secured to the boards in a manner to prevent turning of the studs when tightening or loosening the nuts used to secure lead connectors. Terminal identification corresponding to the lead identifications, specified in Standard MIL-STD-195 shall be permanently marked on the front of the board.

3.2 D.c. machine. - The d.c. machine when operating as a motor on a supply voltage range between 210 to 355 volts d.c. shall be capable of driving a 300 kw. a.c. generator at rated speed and load. When operating as a generator the machine shall be capable of charging a battery over a voltage range of 270 to 355 volts d.c. The d.c. machine shall be in accordance with Specifications MIL-G-3111 and MIL-M-17413 and the following classification:

Rating - Capable of driving an a.c. generator to provide 300 kw. output when the d.c. motor is operating at 210 volts d.c. for 3-1/2 hours and 450 kw. for 5 minutes (10 degree vise will be permitted for the 450 kw. overload in excess of limits specified for rated condition).

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Voltage - 210 to 355 volts d. c.
Insulation - Class B.
Winding - Shunt or compensated.
R. p. m. - Either 1200 or 1800 revolutions per minute (r. p. m.), as determined by the contractor.

3.3 A. c. machine. - The a. c. machine shall operate as a generator when driven by the d. c. machine or shall operate as a synchronous motor when driving the d. c. machine. The a. c. machine shall be in accordance with Specifications MIL-G-3124 and MIL-M-17060 and the following classification:

Rating - 300 kw. continuous and 450 kw. for 5 minutes when operating as a generator.
Voltage - 450 a. c. as generator, 440 volts as a motor.
Phase - 3.
Cycle - 60.
Power factor - 0. 8.
Insulation - Class B.
R. p. m. - As specified in 3. 2.

3.4 Excitation for the a. c. machine. - Except as otherwise specified herein, the excitation shall be in accordance with Specification MIL-G-3124 and MIL-R-2729 and shall be one of the following:

(a) Rotary amplifier use with static magnetic amplifier voltage regulator. - The rotary amplifier exciter shall be either built on common shaft integral with motor-generator or belt driven from motor-generator shaft. The class of insulation shall be class B.
(b) Static exciter and static magnetic amplifier voltage regulator. - The power for the static may be furnished wholly from the rectified a. c. generator output or partially from the d. c. motor supply and the rectified a. c. generator output.

3.5 Excitation for the d. c. machine. - The d. c. machine shall have basic shunt field excited from a 210 to 355 volt d. c. source. In addition, excitation shall be provided to one or more auxiliary fields from the d. c. regulator as required to meet the regulation requirements specified herein.

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3.6 Motor controller. - The motor-generator shall be designed to start from either the a. c. power source or d. c. power source. However, starters shall be provided to start the machine from the d. c. power source only. The motor shall start on no load with base field across the line. The controller shall be in accordance with Specification MIL-C-2174 and the following:

Voltage rating - 250 volts (nominal) d. c. - Operating rating 210 to 335 volts d. c.
Duty - Continuous.
Enclosure - Dripproof.
Operation - Manual and electric remote.
Type - Resistor.
Performance - Nonautomatic and semi-automatic.
Function - Motor starting.
Size - Special - As required.
Protective features - None.
Starting resistance - The controller shall be furnished with starting resistance sufficient to limit inrush currents to 200 percent rated current. The starting cycle shall be 10 seconds.
Voltage variation - The controller shall be designed to operate between 190 and 355 volts d. c. Contactors shall be designed to pick up and seal at 190 volts d. c.
Auxiliary contacts - A minimum of four auxiliary contacts rated each at 10 amperes, 355 volts d. c. or 450 volts a. c. shall be provided in addition to those required by the controller. Two contacts operated by the last accelerating contactor shall be normally open. One contact operated by the line contactor shall be normally open and one normally closed.
Manual operation - The controller shall be so designed that it can be manually operated by one operator to duplicate its electric performance.
Master switches - Remote push button master switches shall be furnished.
Power connection - The controller shall be arranged for the use of Navy type SSGA-800 cable in accordance with Specification MIL-C-2194. Two cables are required for each side of each circuit. The line and load terminals shall be suitable for the use of solderless connectors, Thomas and Belts Bulletin 67, two hole offset tongue, number 51546 or equal. Two connectors required for each terminal will be provided by the installing activity.

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3.7 Overspeed trip device. - An overspeed device shall be installed on the shaft of the motor-generators to prevent it from operating at excessive speeds. Speeds higher than 125 percent of rated speeds shall be considered excessive.

3.8 A. c. voltage regulator. - The a. c. voltage regulator shall be either type II in accordance with Specification MIL-R-2729 for a rotary amplifier system using a magnetic amplifier or type III of Specification MIL-R-2729 for combined static exciter and voltage regulator system except as otherwise specified herein:

(a) Circuit controlled.- Generator field.
(b) Number of regulator.- One per motor-generator (no standby).
(c) Regulation.- The voltage regulator shall be capable of maintaining mean voltage for 3 phase generators with plus or minus 1 percent of rated under any or all of the following conditions:
    (1) Load varying between no load and rated load.
    (2) Load power factor varying between 0.8 lagging and unity.
    (3) Variation of generator frequency.
    (4) Variation of input voltage within specified range.
(d) Regulated voltage adjustment control. - A variable resistor conforming to Specification MIL-R-22 may be used in lieu of a rheostat for the regulated voltage adjustment control. The range of adjustability shall be as specified in Specification MIL-R-2729.
(e) Switching. - The manual automatic switch shall be in accordance with Specification MIL-S-15291. The switching of the voltage regulator from automatic to manual shall be such that the following conditions are met when in manual operation.
    (1) Disconnect (or short out as necessary) all power inputs to the regulator.
    (2) Isolate the regulator to prevent grounding of the motor or generator bus except that potential transformer primary windings need only be disconnected.
(f) Compensation for parallel operation. - Individual droop compensation shall be provided for parallel operation.
(g) Rectifiers. - Rectifiers used in the regulators shall conform to 3.9.5.5.
(h) Capacitors. - Capacitors used in the regulators shall conform to 3.9.5.6.

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3.9 D.c. regulator.- The d.c. regulator shall be a static type (magnetic amplifier) and shall be in accordance with Specification MIL-E-917. The d.c. regulator shall be suitable for controlling the speed of the d.c. machine when operating as a motor or controlling the voltage of the d.c. machine when operating as a generator. The performance of the d.c. machine shall be as specified herein.

3.9.1 Speed regulation.- When the d.c. machine is operating as a motor over the voltage range of 210 to 355 volts d.c. and from no load to full load, the d.c. regulator shall maintain the speed of the motor between 0 and minus 2.5 percent of the rated speed. There shall be no rise in the speed curve from no load to full load at any given input voltage.

3.9.1.1 Regulated speed adjustment.- Provision shall be made to adjust the regulated speed of the motor within the range of plus or minus 2 percent of rated speed in steps of not more than 0.5 percent at any load from no load to full load and within the range of input voltage specified herein.

3.9.1.2 Transient speed response.- With the d.c. motor - operating on voltage range of 210 to 355 volts d.c. and with application of 80 percent kw. load on the a.c. generator in one step from no load, the speed shall recover to within 1 percent of final value within 2 seconds. Under these condition the maximum frequency deviation shall not exceed 7 percent. This value of recovery time and frequency deviation shall be met also upon removal of 50 percent load in one step from 100 percent load. The power factor of the loads shall be approximately 0.8 lagging.

3.9.2 D.c. voltage regulation.- The d.c. voltage regulator shall provide means for adjusting the voltage between 210 and 355 volts d.c. when the d.c. machine is operating as a generator in steps not greater than 1-1/2 volts. It shall be capable of holding any adjusted voltage between 250 and 355 volts within 5 volt band with the speed varying plus or minus 2 percent from rated speed. This 5 volt band regulation applies to single set operation as well as parallel operation. With sudden removal of 150 kw. of d.c. load the maximum voltage rise shall not exceed 10 volts over the preset value and the voltage shall return and remain within 5 volts of the preset value within 1.5 seconds.

3.9.3 Manual adjustment of speed and voltage.- Provision shall be made for manual adjustment of the speed and voltage of the d.c. machine by controlling the base shunt field or control field whichever will result in one of the smaller rheostat. Adjustable resistors in conjunction with manual field rheostat will be permitted. Only one rheostat will be used for controlling speed or voltage. The manual rheostat shall be capable of adjusting the d.c. generated voltage within a range of 250 to 355 volts in steps not greater than 1-1/2 volts. This rheostat shall also be capable of adjusting the speed of the motor within a range of plus or minus 5 percent of rated speed in steps not more than 0.5 percent at any load from no load to full load within the range of input voltage specified herein.

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3.9.4 Automatic transfer controls. - Controls shall be provided integral with the d.c. regulator for automatically transferring the operation of the motor-generator from d.c. generator to a.c. generator and vice versa. When the a.c. machine is operated as a synchronous motor and upon an external signal of the opening of certain circuit breaker the control shall automatically transfer the operation from an a.c. motor to an a.c. generator. Upon receiving an opposite signal from the external source the control shall automatically transfer the operation back to a.c. motor drive.

3.9.5 D.c. regulator construction. -

3.9.5.1 Temperature limits. - The d.c. regulator shall not exceed the temperature rises specified in table I based on rated continuous load and an ambient temperature of 50° Centigrade (C.). The temperature rises of other components shall be as specified in the applicable equipment specification.

Table I - Maximum temperature rises.

Item	Name of part	Temperature rises in °C.
1	Bolted connections and terminal studs:	
1a	Not plated	50
1b	Silver plated	65
2	Coils:	
	Single layer series coils:	
2a	Class A insulation	45
2b	Class B insulation (measured by thermocouple on top of insulation)	65
	Shunt coils:	
2c	Class A insulation	50
2d	Class B insulation	70
2e	Class H insulation (measured by resistance)[1]	130
2f	Class A insulation	40
2g	Class B insulation	60
2h	Class H insulation (measured by thermometer)	110

[1]Preferable method.

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3.9.5.2 Insulation resistance.- The insulation resistance of the d.c. regulator at normal operation temperatures shall be not less than 0.5 megohm (see 4.2.4.3.4).

3.9.5.3 Dielectric strength.- The d.c. regulator shall withstand the dielectric test voltage specified in 4.2.4.3.5.

3.9.5.4 Rheostats.- The manual control rheostat shall be in accordance with Specification MIL-R-15109. The rheostat shall be suitable for back of panel mounting, front of panel operation. A variable resistor conforming to Specification MIL-R-22 may be used in lieu of a rheostat for the regulated frequency adjustment control.

3.9.5.5 Rectifiers.- Rectifiers shall be of the selenium or copper oxide dry disk type in accordance with Specification MIL-R-18281 except that exception may be taken to the qualification approval requirements of Specification MIL-R-18281. However, only those rectifiers which are presently under qualification tests or for which qualification tests have been requested by submittal of all data required by the specification will be acceptable. The maximum cell temperature in the application of rectifier plates shall be limited to 75°C. The cell temperature of a cell near the center of each stack used in the speed and voltage regulators shall be recorded during the full load heat run.

3.9.5.6 Capacitors.- Capacitors for d.c. circuits shall be in accordance with Specification MIL-C-25. Capacitors for a.c. circuits shall be in accordance with either of the following:

(a) Specification MIL-C-25, except that the d.c. rating shall be reduced for a.c. as approved by the bureau or agency concerned.

(b) As approved by the bureau or agency concerned and in accordance with the requirements of section 3 of Specification MIL-C-25, except that the following paragraphs do not apply: Qualification, Dissipation factor, Life, Marking and Specification sheet for individual capacitor types.

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3.9.5.7 Reactors and transformer other than instrument transformers. - Materials and methods of manufacture shall conform to Specification MIL-E-917 and table II.

Table II - Minimum material requirements.

Item	Limitation	Material	Remarks
Electrical insulation as follows:	All sizes	Class A, B, or H	
Conductor	All sizes	As required	Spec. JAN-W-583 or MIL-W-16072
Ground	All sizes	Mica-glass	Spec. MIL-I-3505
Layer	All sizes	Varnished glass	Spec. MIL-I-17205
Tape	All sizes	Woven glass	Spec. MIL-G-1140
Typing cord	All sizes	Glass	Spec. MIL-G-1140
Cable support, Terminal board, Coil spacers, Core spacers	Where used	Glass melamine or glass silicone	Spec. MIL-P-15037, or MIL-P-17721 (class B) MIL-P-997 (class H)
Varnish for treating and impregnating	All sizes	Organic-clear, baking or silicone clear baking	Spec. MIL-V-1137 silicone varnish as approved by the bureau or agency concerned.

3.9.5.8 Switches. - Control switches shall be in accordance with Specification MIL-S-15291. The switching of the d. c. regulator from automatic to manual shall be such that the following conditions are met when in manual operation:

(a) Disconnect (or short out as necessary) all power inputs to the d. c. regulator.
(b) Isolate the d. c. regulator to prevent grounding of the motor or generator bus, except that potential transformer primary windings need only be disconnected.

3.9.5.9 Relays. - The use of relays shall be kept to absolute minimum. Where relays are used they shall conform to Specification MIL-R-2033.

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3.10 Resistors. - Resistors shall conform to Specification MIL-R-15109 except as otherwise approved by bureau or agency concerned.

3.11 Switchboard mounted components. - The following components used in conjunction with a. c. and d. c. regulators and controls shall be mounted remote to regulator enclosures and shall be suitable for switchboard mounting:

(a) A. c. and d. c. regulator adjusting rheostats.
(b) Manual a. c. voltage adjusting rheostat.
(c) Manual d. c. voltage (speed) adjusting rheostat.
(d) Transfer switches.
(e) Current transformers.
(f) Stop and start push buttons.

3.12 Circuit breakers or meters. - Circuit breakers, meters, or synchronizing equipment shall not be furnished as part of the equipment.

3.13 Onboard repair parts. - Onboard repair parts for the motor-generators and associated control equipment shall be in accordance with the applicable specification for each component part. However, no repair armatures, field coils, or armature rewind material shall be furnished. One resistor, capacitor, and rectifier of each type used in the d. c. regulator shall be furnished in each set of onboard repair parts. One of each size of transfer switch shall also be furnished in each set of onboard repair parts. Standard Navy stock numbers shall be obtained for all repair parts, whether furnished or not, in accordance with the procedure specified in Specification MIL-R-15137, except that the reproducible copy of NAVSHIPS 4372 and 4372A shall be submitted within 10 days after all equipment has been released for manufacture. A single copy of NAVSHIPS 4372 with additional copies of NAVSHIPS 4372A as necessary shall be submitted and list all replacement parts for the complete power supply. Submission of separate NAVSHIPS 4372 forms for each major component or piecemeal submission of any of these forms will not be acceptable.

3.14 Drawings. - Master drawings shall be type I in accordance with Specification MIL-D-965 and shall contain information required by the applicable equipment specifications. The d. c. regulator drawing shall contain the same information as required for voltage regulators by Specification MIL-R-2729. In addition the contractor shall furnish an installation drawing for the motor-generator and control equipment. These drawings shall conform to type II of Specification MIL-D-963. The drawings shall show all the information necessary for installing the equipment to include weights, overall dimensions (including clearance for removing parts), foundation bolting details, wiring details

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and center of gravity. A photographic tracing of the approved installation drawing shall be forwarded to the bureau or agency concerned prior to delivery of equipment. Certification data supporting all master drawings shall be submitted and contain all information required by applicable equipment specifications and conform to size A in accordance with Drawing S0103-73729.

3.15 Technical manuals. - Technical manuals shall conform to type B of Specification MIL-T-15071. Two manuals shall be packaged with each motor-generator. The arrangement of the text shall be as follows:

Cover
Title page
General data
    Safety notices
    Component list
        Motor-generator
        Control equipment
            Voltage regulator
            D. c. regulator
            Motor controller
            Motor field rheostat
            Generator field rheostat
Table of contents
List of illustrations and plans
Introduction
    General description
    Power supply requirements and output
Detailed description
    Motor-generator
    Control equipment
        Voltage regulator
        D. c. regulator
        Motor controller
        Rheostats

(Note. - Reference to the motor controller, and rheostats in the appendix is sufficient.)

Installation instructions
    Precautions
    Preparation for use after storage
    Mounting
    Electrical connections
Adjustments and tests

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Principles of operation
Motor generator
Control equipment
Basic magnetic amplifier principles
Voltage regulator
D. c. regulator
Motor-controller (reference to the controller insert is sufficient)
Operating instructions
Maintenance
Parts identification
Test data (include a complete set of combined unit test data not shown on master drawings)
Drawings
Appendix
Motor controller insert
Rheostat insert

3.16 Reports. -

3.16.1 Progress reports. - A report shall be submitted bi-weekly, giving details of significant progress during the month. The first report shall include schedules of estimated completion dates for various phases of the programs for all component parts, such as basic design work, preparation of detail drawings, assembly, testing, and shipment. Subsequent reports shall indicate the percentage completion of the work under each of these items.

3.16.2 Test reports. - A complete set of test data shall be forwarded to the bureau or agency concerned for approval prior to shipment of equipment.

3.17 When any specification which forms a part of this specification requires that the product be subjected to and pass qualification tests, only products which are listed on the applicable Qualified Products List on the date of invitation for bids or which may be added to that Qualified Products List subsequent to that date shall be utilized in the construction of equipment specified to be in accordance with this specification. In the event no Qualified Products List has been issued, the contractor shall request instructions as to what testing will be required to determine whether the product meets the requirements of the specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection procedures. - For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

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4.2 Tests.- Sampling, inspection and tests shall be in accordance with each individual specification and as specified herein.

4.2.1 Motor-generator tests.- Motor-generators shall be tested in accordance with Specifications MIL-G-3111, MIL-G-3124, MIL-M-17060 and MIL-M-17413 as applicable and as follows:

4.2.1.1 Heating.- The heat run tests shall be conducted with each of the a.c. and d.c. machines operating as a motor and as a generator. The heat run on the d.c. machine shall be conducted at the high and low voltage limit of the voltage range specified herein. All heat runs shall be conducted with cooling water inlet temperature at 85° F.

4.2.1.2 Inclined operation.- The motor generator shall be subjected to the following inclination tests:

(a) A.c. end inclined 30 degrees.
(b) D.c. end inclined 30 degrees.
(c) Shaft horizontal base tilted 15 degrees to right.
(d) Shaft horizontal, base tilted 15 degrees to the left.

In each of the positions specified herein the equipment shall be energized and operated for 30 minutes. During this test it shall be ascertained that there is no sudden increase in bearing temperatures, evidence of oil or grease leakage either external or internal or increase in noise of the motor generator.

4.2.1.3 Enclosure test for motor-generator.- The motor-generator enclosure shall be tested as follows:

(a) The enclosure of the first completed motor-generator without armature installed but with field coils installed, shall be immersed in water to within 1/4 inch of the bottom of the shaft opening for a period of 24 hours. There shall be no leakage into the motor-generator.
(b) The frame of the first completely assembled motor-generator shall be subjected to a stream of water from a hose having a 1-inch nozzle diameter, under a head of 55 feet from a distance of 10 feet. The stream of water shall be directed to successfully cover all external surfaces of the motor-generator for a total period of not less than 5 minutes. There shall be no leakage into the motor-generator except around the shaft openings.

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4.2.1.4 Water-air-cooler test.- The water-air-cooler shall be tested in accordance with Specification MIL-E-2036 except that the hydrostatic tests shall be made at a test pressure of 150 percent of the rated cooler operating pressure.

4.2.1.5 Audio noise test.- The motor-generator shall be tested to determine compliance with 3.1.7. The noise level shall be measured using the flat network and weighing network for 70 db in accordance with American Standard Z24.3. Measurements shall be made in accordance with procedure specified in American Standard Z24.7, except for the following:

(a) No observers or reflecting surfaces, other than the floor when necessary, should be less than 3 feet from either the microphone or the equipment being measured.
(b) In cases where the measurements cover a range of 10 db or more, or in the case of apparatus producing a highly directional noise, the individual ratings at each microphone location shall be reported and summarized in terms of power average together with the maximum and minimum values.

4.2.2 A.c. voltage regulator test.- Tests shall be conducted in accordance with Specification MIL-R-2728 except as follows:

(a) Combined unit test (motor-generator and regulator) shall be conducted with the d.c. motor operating at the high and low voltage limits of the voltage range specified herein.
(b) Vibration tests shall be conducted in accordance with 4.2.4.3.14 as specified herein.
(c) Inclination test shall be conducted in accordance with 4.2.4.3.13 as specified herein.

4.2.3 Motor controller tests.- Motor controller shall be tested in accordance with Specification MIL-C-2174 except as follows:

(a) Vibration tests shall be conducted in accordance with 4.2.4.3.14 as specified herein.
(b) Inclined operation tests shall be conducted in accordance with 4.2.4.3.13 as specified herein.

4.2.4 D.c. regulator tests.- D.c. regulators shall be tested at the manufacturer's plant in the presence of the Government inspector. Those tests required to be conducted on the regulator when operating with the motor-generators shall be conducted at the prime contractors plant. In order to prevent needless retesting, the manufacturer shall make sufficient preliminary tests, previous to

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in MIL-C-24105 unless otherwise specified

tests to be witnessed by the Government inspector, to insure that the design of the equipment conforms in all respects to the requirements of this specification. To prevent delays, not more than two tests shall be made. The second shall be made within such time after the first as stipulated by the Government inspector. Failure to make the necessary repair or remedy the defects within that time shall be considered sufficient cause for rejection of the part. Each part or complete d.c. regulator shall be subjected to the tests specified in table III.

Table III - Tests.

Description of test	First unit test	Routine tests
Material	4.2.4.3.1	4.2.4.3.1
Adjustment and fit	4.2.4.3.2	4.2.4.3.2
Resistance and capacitance measurements	4.2.4.3.3	4.2.4.3.3
Insulation resistance	4.2.4.3.4	4.2.4.3.4
Dielectric	4.2.4.3.5	4.2.4.3.5
Class HI shock	4.2.4.3.6	---
Heating	4.2.4.3.7	---
Creepage and clearance	4.2.4.3.8	---
Speed regulation	4.2.4.3.9	---
Voltage regulation	4.2.4.3.10	---
Recovery	4.2.4.3.11	---
Enclosure	4.2.4.3.12	---
Inclined operation	4.2.4.3.13	---
Vibration	4.2.4.3.14	---
Parallel operation	4.2.4.3.15	---
Speed response	4.2.4.3.9.1	---

4.2.4.1 First unit tests.- The first unit or part thereof furnished on a contract or order shall be subjected to the first unit tests specified in table III. Nonconformance will result in rejection of the design. If the design of any unit or part is changed after the first tests have been performed, the decision shall be made by the Government inspector as to whether a retest will be required.

4.2.4.2 Routine tests.- Each d.c. regulator shall be subjected by or under the supervision of the Government inspector to the routine tests specified in table III to determine conformance with the requirements of this specification. Nonconforming d.c. regulators shall be individually rejected.

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4.2.4.3 Methods and tests. -

4.2.4.3.1 Material tests. - While it is not the intention of this specification, in general, to require that all the material used in the construction of the control equipment be tested in accordance with Government specifications referred to in each individual case, the Government inspector shall require such material tests whenever, in his judgement, it is necessary to ascertain that the quality of the material used is at least equal to the material specified herein, and covered by the referenced specification, or as shown on manufacturer's approved drawings.

4.2.4.3.2 Adjustment and fit. - All moving parts of the regulator shall be examined to ascertain that they move freely and without unintentional looseness. The adjustment of all contactors or relays shall be examined to determine that they are within the correct tolerances as shown on approved drawings.

4.2.4.3.3 Resistance and capacitance measurements. - The resistance of all resistors and coils and capacitance of all capacitors shall be accurately measured at room temperature (approximately 20° C.) and at 60 cycle frequency to determine that the resistance or capacitance is within plus or minus 10 percent of the design value. The temperature at which the measurements are made shall be recorded from thermometers in air and oil.

4.2.4.3.4 Insulation resistance. - The measurement of insulation resistance shall be made with all circuits of equal voltage above ground connected together. Circuits or groups of circuits of different voltage above ground shall be tested separately. Tests shall be made using a potential of 500 volts d.c. The temperature of the windings at the time of the insulation resistance measurement shall be recorded.

4.2.4.3.5 Dielectric test. - The dielectric test shall be made after all tests have been completed. The test shall be made upon the completely assembled unit and not upon an individual part; and exception is made in the case of repair parts which require dielectric tests, for example, repair coils. The test voltage shall be applied both between circuits and between each circuit and ground. The dielectric test voltage shall be twice the rated voltage plus 1,000 volts but not less than 1,500 volts applied for a period of 1 minute. The frequency of the test voltage shall be not less than 60 cycles and shall be approximately a true sine wave.

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4.2.4.3.6 High-impact shock test. - The test shall be as specified in Specification MIL-S-901. The features of the test shall be as follows:

(a) The required type of shock test and weight designation of the shock test. - As required by the complete control panel.

(b) A definition of "failure to perform principal function".

(1) Breakage of any parts, including mounting bolts. Cracking or signs of cracking in parts vital to operation shall be considered breakage.

(2) Appreciable distortion or dislocation of any parts such as mounting feet, coils, or conducting terminals.

(3) A value of insulation resistance (corrected to 25° C.) less than that permitted by 3.9.5.2.

(4) Failure to withstand dielectric test voltage equal to 65 percent of that specified in 3.9.5.3.

(5) Failure to maintain the sensitivity specified in 3.9.1 and 3.9.2. The test shall be conducted with the regulator controlling the machine at any convenient load. The regulation of the machine shall be recorded before and after each blow. No adjustment shall be made between blows.

(c) Acceptable method of mounting. - The regulator shall be shock-tested within the control panel and mounted on the shock machine in a specified manner which simulated typical shipboard installations.

(d) Place at which shock test will be conducted. - Where a manufacturer is unable to conduct these tests at his own plant, he may arrange to have them conducted at a commercial laboratory approved by the bureau or agency concerned.

(e) Disposal of shock tested regulator.

(1) Regulators which have been subjected to the high-impact shock test and have failed to conform to the requirements will not be acceptable, either in whole or any of the parts.

(2) Regulators which have been subjected to the high-impact shock tests and have successfully passed this test may apply on a contract or order, provided the following post-shock tests are conducted and requirements satisfactorily met:

a. Regulation.
b. Transient response.
c. Heating.
d. Dielectric.
e. Insulation resistance.

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4.2.4.3.7 Heating.- Heating tests shall be made on the regulator equipment when operated under rated voltage and current conditions. The thermometer, thermocouple, or resistance method shall be used to determine the temperature rise on representative current carrying parts such as coils, reactors, transformers, resistors, rectifiers and terminals. Heating tests shall be made using the intended enclosure, with each item in its respective place. The maximum permissible temperature rise over a convenient room ambient (not less than 10° C.) shall not exceed the values specified in table I.

4.2.4.3.8 Creepage clearance distances shall be demonstrated by actual measurement to be in accordance with Specification MIL-E-917.

4.2.4.3.9 Speed regulation.- A total of 5 test runs shall be conducted to determine compliance with 3.9.1. With 210 volts d.c. applied to the motor, the generator load shall be varied from no load to full load at rated power factor in approximately 5 equal steps and back to no load in the same manner. These tests shall be repeated for unity power factor. Each of these tests shall be repeated with 220, 250, 280, and 355 volts applied to the motor. The adjustment shall be made on the speed governor between or during tests when motor is operating on voltages between 220 and 280. Initial adjustment at any other voltages will be permitted. The frequencies recorded by this test shall be within limits specified in 3.9.1.

4.3.4.3.9.1 Speed response.- With the d.c. motor operating on 210 volts d.c. and a.c. generator at no load, apply 80 percent a.c. load at 0.8 power factor in one step. With the d.c. motor operating at 210 volts and a.c. generator at 100 percent load and 0.8 power factor the a.c. load shall be reduced in one step to 50 percent load at 0.8 power factor. Oscillograms shall be taken to determine compliance with 3.9.1.2. The test shall be repeated with the d.c. motor operating on 355 volts d.c.

4.2.4.3.10 Voltage regulation.- The d.c. generator employed in this test shall be loaded with the regulator operating to control the generator voltage. The output voltage of the d.c. generator shall be adjusted for 250 volts. The load of the d.c. generator shall be varied from no load to full in five approximately equal steps and back to no load in same manner. This test shall be repeated for for 280 and 355 volts. With the same regulator setting the above tests shall be repeated at 2 percent above and below rated speed. The voltages shall be read for each of the above load points to determine compliance with 3.9.2. The voltage shall be adjusted between 210 and 355 volts by means of the regulator adjusting rheostat to determine compliance with 3.9.2.

4.2.4.3.11 Recovery.- With the d.c. generator operating at rated speed, rated load and at 250 volts, a load of 140 kw. shall be suddenly removed without any adjustment to speed or voltage. The voltage shall be measured by means of an oscillograph to determine compliance with 3.9.2. This tests shall be repeated for 350 volts.

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4.2.4.3.12 Enclosure.- Visual inspection shall be made for compliance with Specification MIL-E-2036.

4.2.4.3.13 Inclined operation.- The equipment or portion thereof shall be operated in an inclined position of 30 degrees to each side for 30 minutes. The equipment shall continue to function during inclined operation.

4.2.4.3.14 Vibration.- Individual parts or subassemblies of control equipment of a new design not previously tested, by the bureau or agency concerned shall be subjected to type I vibration testing in accordance with Standard MIL-STD-167.

4.2.4.3.15 Parallel test.- With the motor generators connected to the same 440 volt a. c. power supply (voltage regulators in automatic operation) the control equipment shall be adjusted so that rated voltage is obtained and the kw. loads of each d. c. generator are equal for a combined load of 20 percent. No further adjustment of the control equipment shall be made for the remainder of the test. The total loads shall then be varied from 20 percent to 100 percent of the kw. rating of the parallel d. c. generators in four approximately equal steps and back to 20 percent load in the same manner. The voltage, current, kw. load, field voltage, and field current of each generator at each load point shall be recorded. The test shall be conducted with the d. c. generator supplying 270 and 355 volts.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking.- Motor generators, control equipment and associated repair parts shall be preserved, packaged, packed and marked for domestic shipment - immediate use, or domestic shipment and storage as specified in the contract or order, in accordance with Specifications MIL-P-16016 and MIL-P-16298, as applicable. All regulators and motor-generators shall be preserved and packaged by method II of Specification MIL-P-116 regardless of size or weight for domestic shipment and storage or overseas shipment.

5.2 Technical manuals.- Technical manuals, pamphlets and handbooks, as applicable shall be packaged and packed in accordance with Specification MIL-T-15071.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the title, number, and date of this specification.

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Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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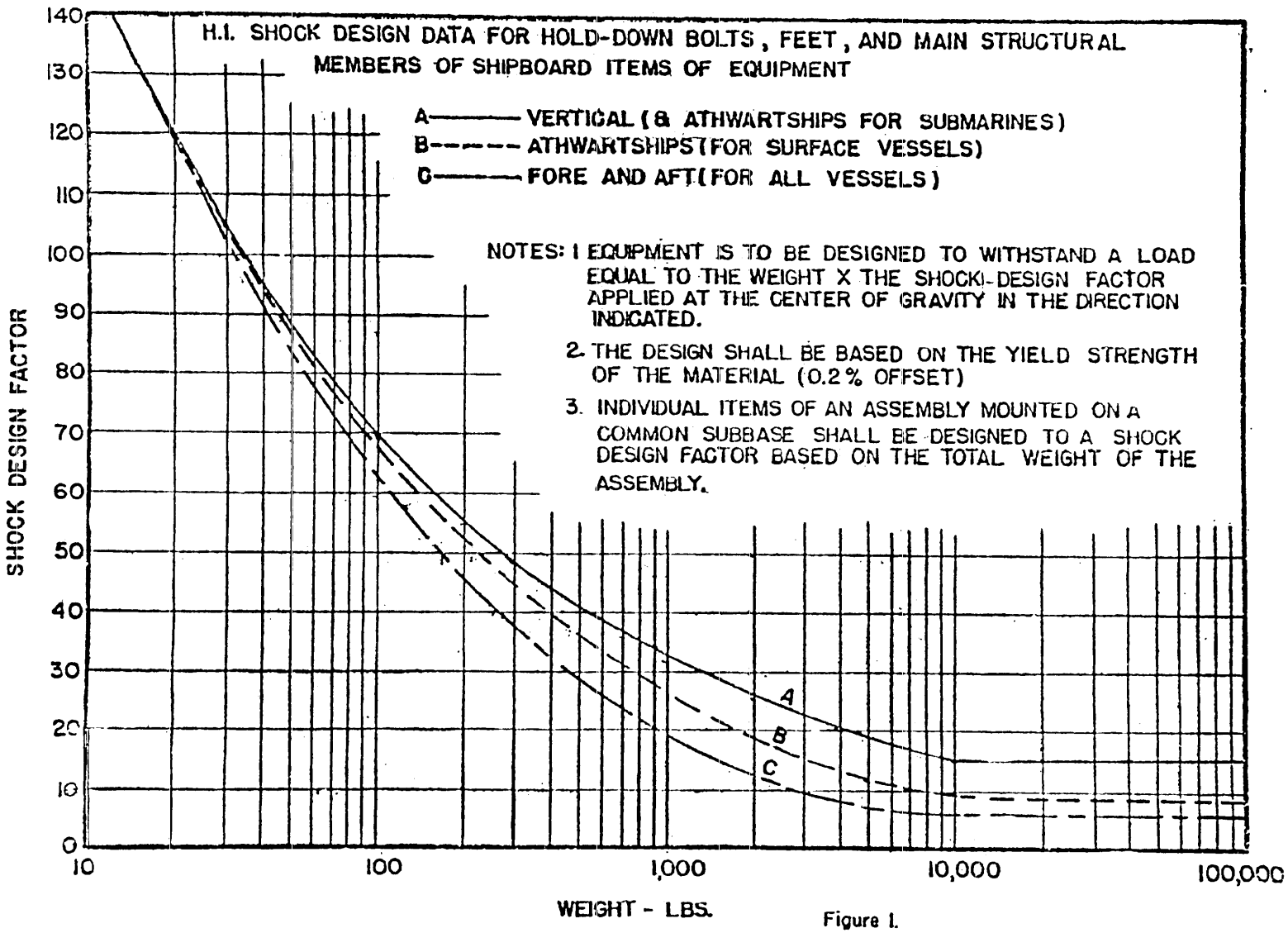


Figure 1.

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